Kubernetes

Kubernetes is an **open-source container orchestration platform**
It used to manage, deploy, and scale containerized applications automatically.

Why Do We Need Kubernetes?

When working with **Docker**, we can run containers easily.

But in real-world production:

- You might have **100+ containers**
- Need **auto-scaling**
- Need **load balancing**
- Need **auto-restart if container crashes**
- Need **rolling updates without downtime**

Managing this manually is impossible.

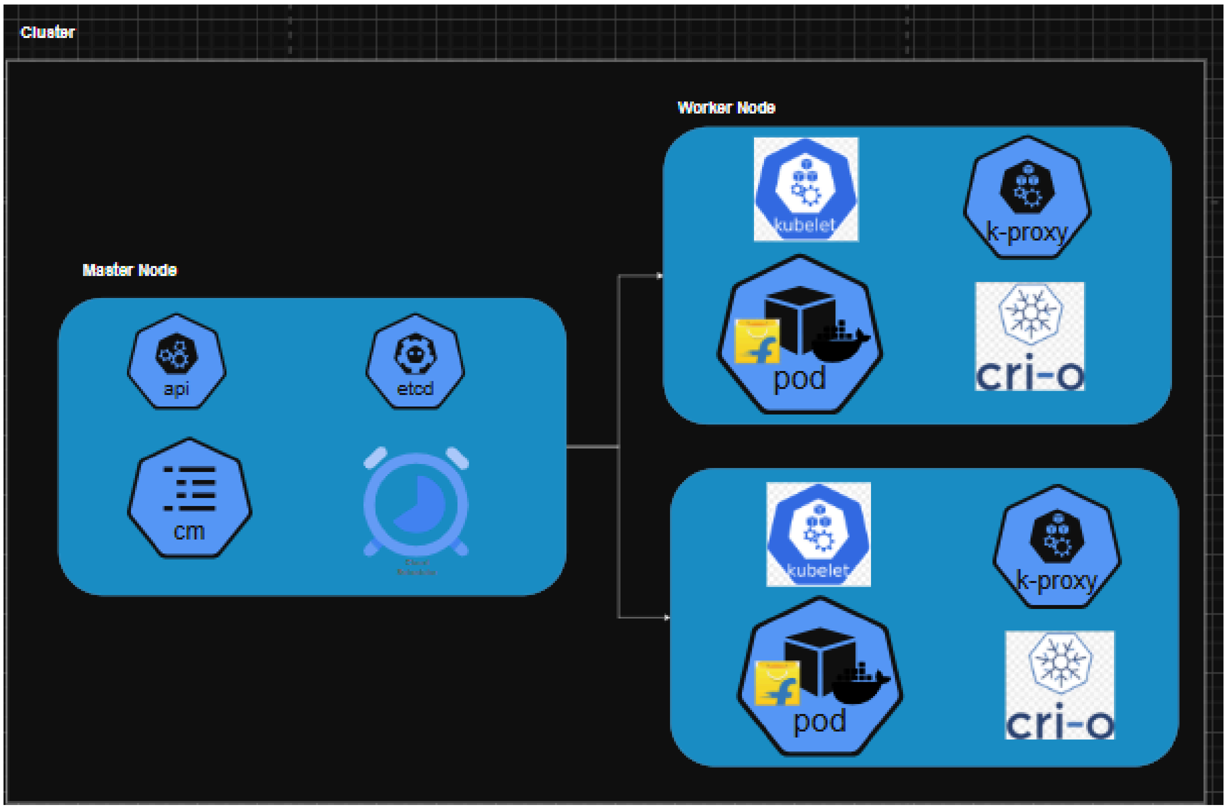
To solve this we need Kubernetes

What Kubernetes Does

1. **Container Orchestration** : Automatically manages multiple containers.
2. **Auto Scaling:** Scales pods up/down based on traffic.
3. **Self-Healing:** If a container crashes → Kubernetes restarts it.
4. **Load Balancing:** Distributes traffic across multiple pods.
5. **Rolling Updates**: Deploy new version without downtime.

Architecture

Kubernetes works in a **cluster**:



Cluster:

A Cluster is a group of machines (nodes) that work together to run containerized applications.

Cluster = Master Node+ Worker Nodes

Master Node

1. kube-api

- Main entry point to Kubernetes
- Communicates with all other components

2. etcd

- A key-value database
- Stores entire cluster state: Pods,Nodes,Configurations,Secrets

3. Controller manager

Continuously check the cluster state and make sure it matches the desired state.

4. Schedullers : Assigns pods to nodes.

Worker Node

- Responsible for running the actual applications

1. kubelet:

- It is an agent running on each worker node.
- It talks with API server
- It ensures container in pod are running
- It will tell the container runtime to start the container image.

2. k-proxy:

- It will contain network info like node ip,pod ip and svc ip.
- It handles the networking , service routing and load balancing between pods.

3.container runtime:

- It will provide the container env to run the environment.
- It is actually runs the container.